5. **Calculator active.** Using the information in the table below, find the average rate of change for each given interval. Include units.

t seconds	24	60	135	200
$s(t)$ miles	8	1.3	3.5	0.3

a. $135 \leq t \leq 200$

b. $24 \leq t \leq 135$

c. $24 \leq t \leq 200$

6. **Calculator active.** Estimate the rate of change of $f(x) = x^2 - x$ at $x = -1$

7. Mr. Gardener is decreasing the amount of water used on his lawn each month, and the height of his grass is decreasing. Does this scenario represent a positive or negative rate of change?

What is the average rate of change for each function on the given intervals?

8. $y = 7 - 2x$ on $-4 \leq x \leq 1$

9. $y = 3x^2 - 2x + 1$ on $-1 \leq x \leq 2$

What is the rate of change of the average rates of change for each function over consecutive equal-length intervals?

10. $y = 5x + 9$

11. $f(x) = 2x - 5x^2$.

12. The values of a function are given at selected x-values in the table below. The function's concavity does not change. Determine if the function is concave up or concave down. Justify your answer.

x	5	9	13	17	21
$g(x)$	45	20	0	-10	-14

Find the leading coefficient and the degree of each polynomial.

13. $f(x) = x^5 - 2x^2$

L.C. _____ Degree: _____

14. $f(x) = 10 - 3x^2 + 7x^3 - 2x$

L.C. _____ Degree: _____

Let $f(x)$ be a polynomial function with the given values. Are there any guaranteed extrema? If so, state where they occur.

15. $f(0) = -3, f(4) = 0$, and $f(7) = 0$.

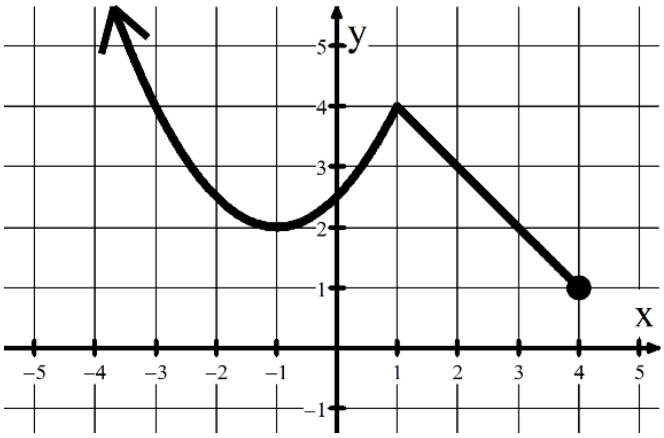
16. $f(-7) = 0, f(-2) = 5, f(0) = 1$, and $f(9) = 0$.

Is there a global maximum or minimum for each function?

17. $f(x) = -5x^6 + 6x^4 - 3x^3 + 1$

18. $f(x) = 2x^5 + x^2 - 6$

Find the following extrema. If there are none, cross it off and write NONE.



19. Absolute **min** of _____ when $x =$

20. Absolute **max** of _____ when $x =$

21. Relative **min**(s) at $x =$

22. Relative **max**(es) at $x =$

For each polynomial function, find the intervals for each condition.

23. $h(x) = x^2 - 8x + 15$. When is $h(x) \leq 0$?

24. $f(x) = -x^3 + 5x^2 + 24x$. When is $f(x) \geq 0$?

25. The degree of a polynomial is 7 with real zeros at $x = -8, x = 1$, and $x = 4$. $x = 1$ has a multiplicity of 3. How many non-real zeros does the polynomial have?

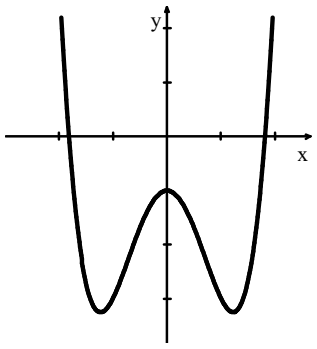
26. $5 - i$ is a non-real zero of a polynomial, find another zero.

27. Find the **degree** of the polynomial from the given input and output values.

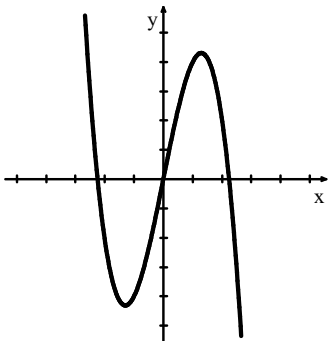
Input	0	1	2	3	4	5	6	7
Output	2	-2	4	20	46	82	128	184

State whether the following graphs represent functions that are even, odd, or neither.

28.



29.



State if the following functions are even, odd, or neither.

30. $f(x) = 4x^7 + 5x^3 - 2x$

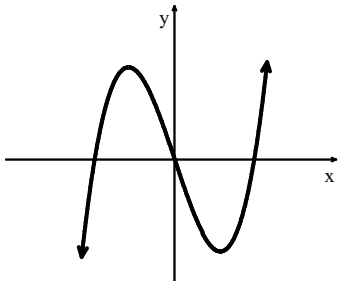
31. $f(x) = 7 - 6x^8 - 3x^2$

Describe the end behavior of each function using limit notation.

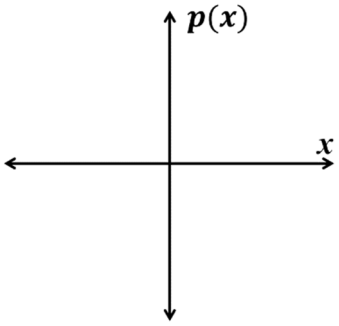
32. $p(x) = -11x^7 - 6x^2 + 4x$

33. $p(x) = -7x^6 + 4x - 8$

34.



35. Sketch the graph of a polynomial function that could match statements $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = \infty$.



Name: _____ Date:_____

Corrective Assignment

Unit 1A CA – Polynomial and Rational Functions

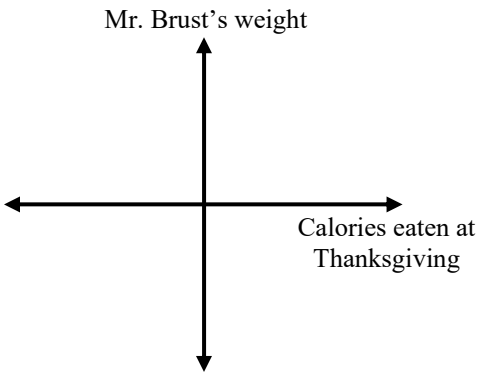
For each function, identify what the dependent and independent variables represent.

1. $r(c)$ is a function where c is the number of coyotes and r is the number of rabbits on the property.

Dep:

Indep:

2.



Dep:

Indep:

3. Let the function f be increasing or decreasing, but not both. State whether the function is increasing or decreasing on the given interval and justify your answer.

x	15	30	50	75	110
$f(x)$	43	40	35	25	12

4.

a. On what interval(s) is the graph concave up?

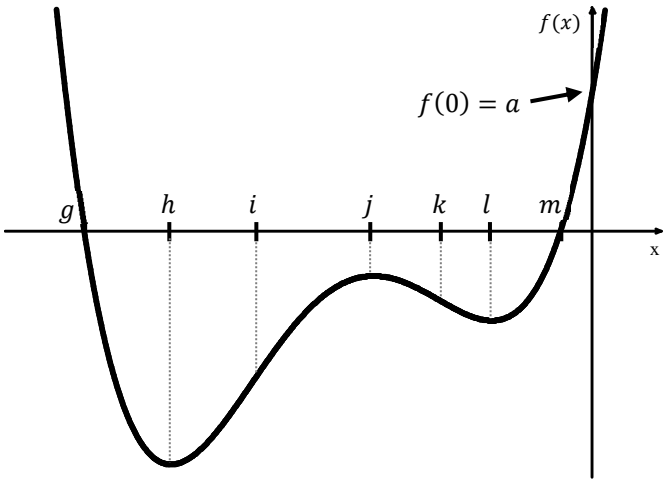
b. On what interval(s) is the graph concave down?

c. On what interval(s) is the graph increasing?

d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.



5. **Calculator active.** Find the average rate of change of the function $h(t) = t + 3^t$ on the interval $\frac{1}{2} \leq x \leq 1$.

6. Using the information in the table below, find the average rate of change for each interval.

t weeks	2	4	6	8
$B(t)$ wild boar	2	12	17	20

a. $2 \leq t \leq 8$

b. $6 \leq t \leq 8$

c. $2 \leq t \leq 6$

7. **Calculator active.** Estimate the rate of change of $f(x) = \frac{1}{x}$ at $x = 2$.

8. Increasing the number of cigarettes someone smokes increases their chance of lung cancer. Does this scenario represent a positive or negative rate of change?

What is the average rate of change for each function on the given intervals?

9. $y = 3x - 6$ on $2 \leq x \leq 5$

10. $y = -x^2 + 3x + 4$ on $-2 \leq x \leq 1$

What is the rate of change of the average rates of change for each function over consecutive equal-length intervals?

11. $y = 7 - 3x$

12. $f(x) = 2x^2 + 3x - 6.$

13. The values of a function are given at selected x-values in the table below. The function's concavity does not change. Determine if the function is concave up or concave down. Justify your answer.

$\boldsymbol{x}$	6	8	10	12	14
$\boldsymbol{g(x)}$	8	6	0	-14	-34

Find the leading coefficient and the degree of each polynomial.

14. $f(x) = 4x^2 - 3$

L.C. _____ Degree: _____

15. $f(x) = -8x^3 + 5 + 9x^4$

L.C. _____ Degree: _____

Let $\boldsymbol{f(x)}$ be a polynomial function with the given values. Are there any guaranteed extrema? If so, state where they occur.

16. $f(-13) = 0, f(0) = 5,$ and $f(2) = 3.$

17. $f(-1) = 0, f(0) = -8, f(4) = -3,$ and $f(6) = 0.$

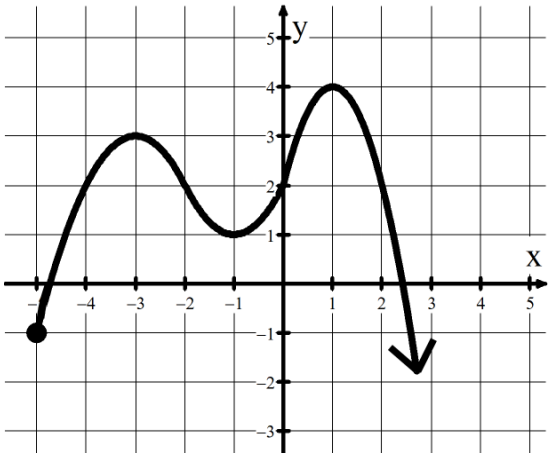
Is there a global maximum or minimum for each function?

18. $f(x) = -2x^7 + 1$

19. $f(x) = 3x^4 - 4x^3 - 2$

20. $f(x) = -2x^6 + 7x^5 - x^3 + x$

Find the following extrema. If there are none, cross it off and write NONE.



21. Absolute **min** of _____ when $x =$

22. Absolute **max** of _____ when $x =$

23. Relative **min**(s) at $x =$

24. Relative **max**(es) at $x =$

For each polynomial function, find the intervals for each condition.

25. $f(x) = -x^2 - 4x + 45$. When is $f(x) \geq 0$?

26. $g(x) = (x+5)^2(x-1)^3(x-6)$. When is $g(x) \geq 0$?

For each polynomial, the degree is listed along with all of its real zeros. Find the number of NON-REAL zeros the polynomial has.

27. The degree of a polynomial is 5 with real zeros at $x = -3, x = 0$, and $x = 2$. $x = 2$ has a multiplicity of 3.

28. The degree of a polynomial is 10 with real zeros at $x = -5, x = -2$, and $x = 11$. $x = 11$ has a multiplicity of 4.

Given one non-real zero of a polynomial, find another zero.

29. $9 + 2i$

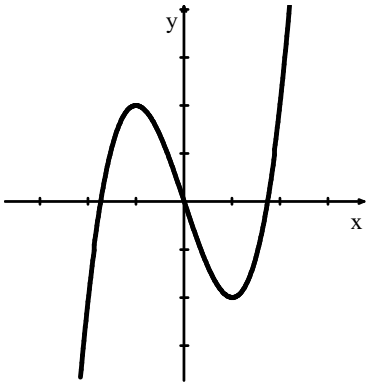
30. $-8 - i$

31. Find the **degree** of the polynomial from the given input and output values.

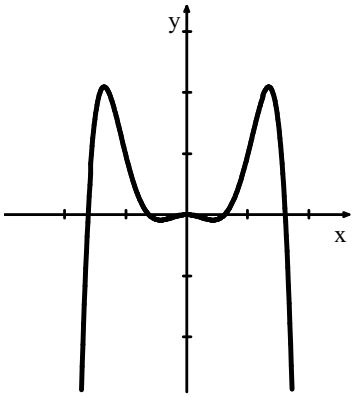
Input	0	1	2	3	4	5	6	7
Output	100	107	118	115	80	-5	-158	-397

State whether the following graphs represent functions that are even, odd, or neither.

32.



33.



State if the following functions are even, odd, or neither.

34. $f(x) = x^4 + x + 1$

35. $f(x) = 6x^5 - 2x^3 + x$

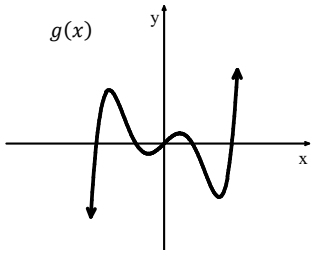
36. $f(x) = 3 + 5x^6 - 3x^2$

Describe the end behavior of each function using limit notation.

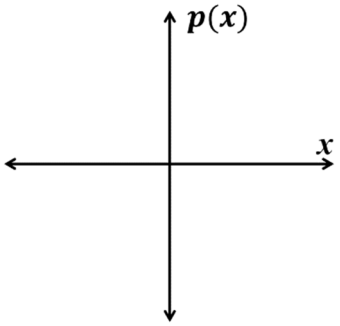
37. $f(x) = 7x^4 - 12$

38. $p(x) = -8x^{13} - 6x^4 + x$

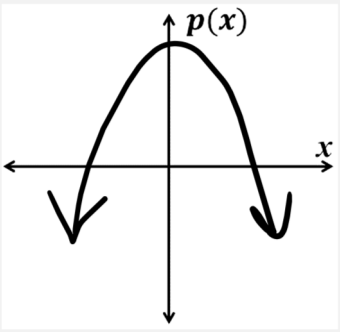
39.



40. Sketch the graph of a polynomial function that could match statements $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = -\infty$.



Answers to Unit 1A Corrective Assignment

1. Dependent: Number of rabbits. Independent: Number of coyotes.			2. Dependent: Mr. Brust's weight. Independent: Calories eaten at Thanksgiving.		
3. f is decreasing on the interval $15 < x < 110$ because for all a and b in the interval, if $a < b$, then $f(a) > f(b)$.			4a. $x < i$ and $x > k$	4b. $i < x < k$	
4c. $h < x < j$ and $x > l$	4d. $x < h$ and $j < x < l$		4e. $x = g$ and $x = m$	4f. $y = a$	
5. 3.5358	6a. 3 wild boar per week	6b. 1.5 wild boar per week	6c. 3.75 wild boar per week		
7. ≈ -0.25	8. positive	9. 3	10. 4	11. 0	12. 4
13. Concave down because the rate of change is decreasing over equal-length input-value intervals.			14a. 4	14b. 2	15a. 9	15b. 4
16. No guarantee.	17. Yes, on $-1 < x < 6$.		18. none	19. min	20. max
21. none	22. max of 4 when $x = 1$		23. -5 and -1	24. -3 and 1	
25. $[-9, 5]$	26. $(-\infty, 1] \cup [6, \infty)$	27. 0	28. 4	29. $9 - 2i$	30. $-8 + i$
31. 3	32. odd	33. even	34. Neither	35. Odd, $f(-x) = -f(x)$	
36. Even, $f(-x) = f(x)$	37. $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = \infty$		38. $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = -\infty$		
39. $\lim_{x\to-\infty} g(x) = -\infty$ and $\lim_{x\to\infty} g(x) = \infty$			40. 		

AP Precalc

Write your questions and thoughts here!

1.7A Rational Functions & End Behavior

1.7A Notes

Rational Function

The ratio of two polynomials where the polynomial in the denominator cannot equal 0.

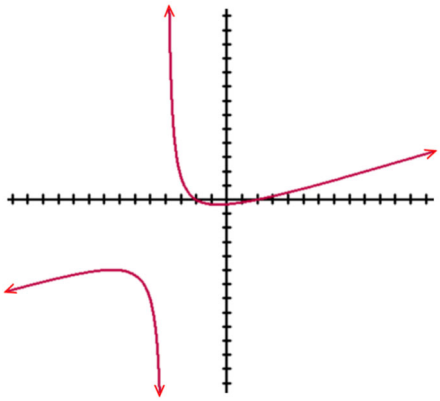
$$r(x) = \frac{f(x)}{g(x)} \text{ where } f \text{ and } g \text{ are polynomials and } g(x) \neq 0$$

Domain of Rational Functions

Example 1:

1. $f(x) = \frac{x^2-4}{3x+12}$

Domain:



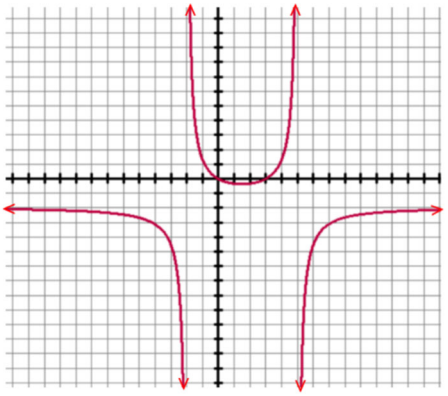
Find the domain of the rational functions.

2. $g(x) = \frac{4x-1}{x^2-2x-24}$

Domain:

3.

Domain:

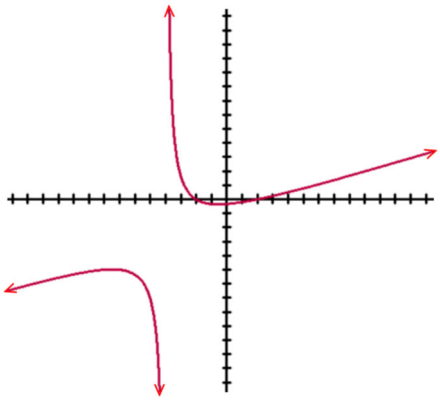


End Behavior of Rational Functions

Example 4:

4. $f(x) = \frac{x^2-4}{3x+12}$

End Behavior:



Write your questions and thoughts here!

Determine the end behavior of the following rational functions.

5. $g(x) = \frac{4x-1}{x^2-2x-24}$

As x increases without bound the $g(x)$…

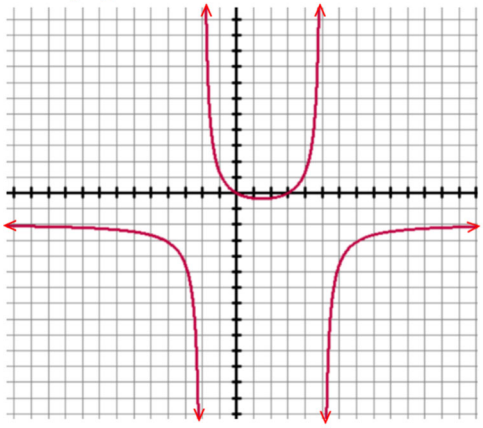
x	$g(x)$
10	
100	
1,000	
10,000	

As x decreases without bound the $g(x)$…

x	$g(x)$
-10	
-100	
$-1{,}000$	
$-10{,}000$	

End Behavior:

6. The graph of f



End Behavior:

Horizontal Asymptote

The line $y = b$ is a horizontal asymptote of the graph of f when…

$$\lim_{x\to-\infty} f(x) = b \qquad \text{OR} \qquad \lim_{x\to\infty} f(x) = b$$

1.7A Rational Functions and End Behavior

AP Precalculus

1.7A Practice

State the domain of the following rational functions. Use interval notation.

1. $f(x) = \frac{x+1}{2x+5}$

Domain:

2. $g(n) = \frac{n^2-4}{(n+3)(n-7)}$

Domain:

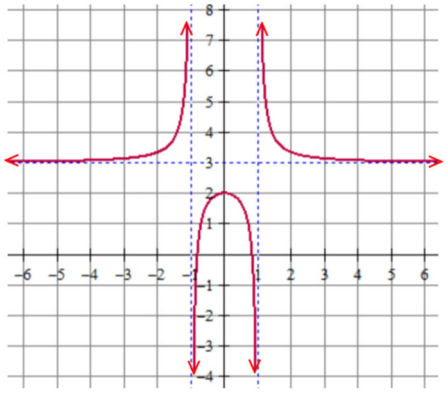
3. $h(x) = \frac{x^3+7x^2+12x}{x^2+2x-15}$

Domain:

State the domain of the following rational functions. Use interval notation.

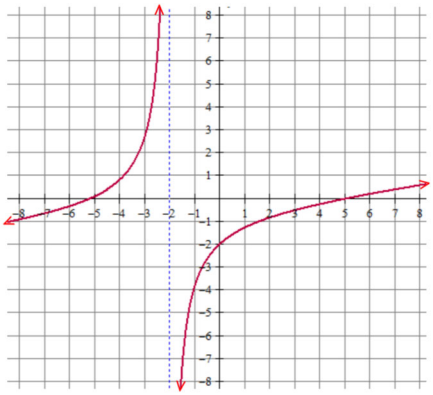
4.

Domain:



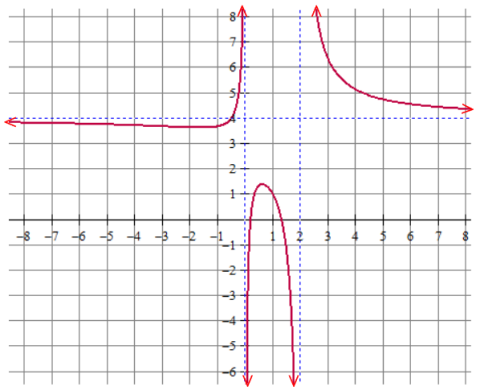
5.

Domain:



Use the graph of the rational function f to find the following.

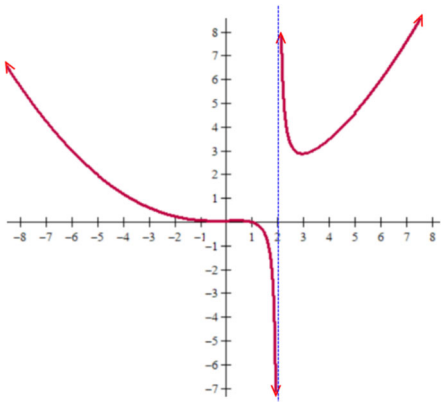
6.



End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

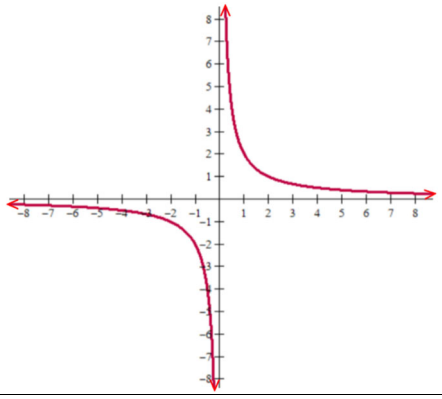
7.



End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

8.



End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

CALCULATOR ACTIVE Complete the table to answer the following.

9. $f(x) = \frac{x^2}{x^2-4x}$

x	$-10{,}000$	$-1{,}000$	-100	100	$1{,}000$	$10{,}000$
$f(x)$						

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

---

10. $d(t) = \frac{t^3-3t}{2t+1}$

t	$-5{,}000$	-500	-50	50	500	$5{,}000$
$d(t)$						

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

Use the table of the rational function *h* to find the following.

11.

t	$h(t)$
$-999{,}999$	4.2×10^{32}
$-99{,}999$	7.3×10^{12}
$-9{,}999$	$6{,}524{,}378$
-999	$561{,}391$
0	32
999	-478
$9{,}999$	$-289{,}437$
$99{,}999$	$-2{,}198{,}235$
$999{,}999$	$-389{,}031{,}455$

a. Find $h(999) =$

b. Find $\lim_{t\to\infty} h(t) =$

c. As *t* increases without bound the $h(t)$…

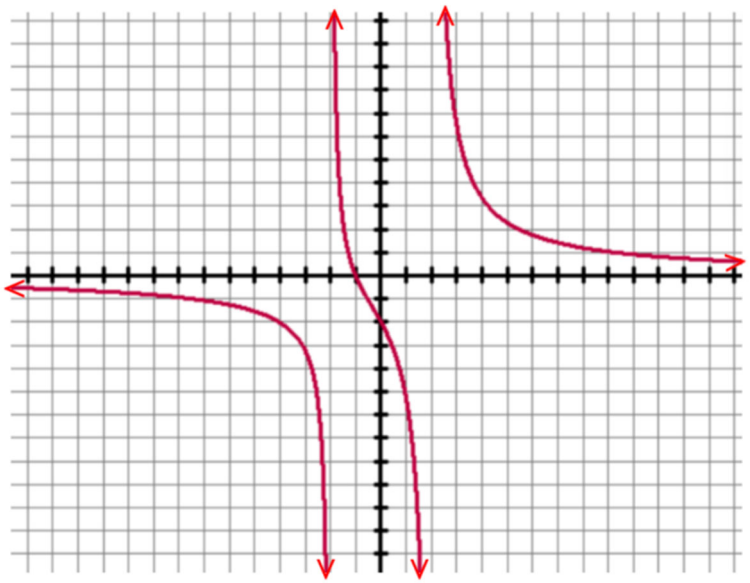
d. Find the *y*-intercept.

e. Find $\lim_{t\to-\infty} h(t) =$

f. As *t* decreases without bound the $h(t)$…

Use the graph of the rational function f to answer the following if they exist!

12.



a. State the domain of $f(x)$.
Use interval notation.

b. Find $f(9) =$

c. Find the y-intercept.

d. Find the x-intercept(s).

e. State the interval(s) where $f(x)$ is increasing.

f. State the interval(s) where $f(x)$ is decreasing.

g. Find $\lim_{x\to\infty} f(x) =$

h. As x increases without bound the $f(x)$…

i. Estimate the point of inflection.

j. Find the average rate of change over the interval $-1 \le x \le 0$.

k. Find $\lim_{x\to-\infty} f(x) =$

l. As x decreases without bound the $f(x)$…

m. Find the horizontal asymptote.

n. State the interval(s) where $f(x) < 0$.

13. Which one of the following polynomials could be the denominator for the graph of the rational function shown above in question #12 ?

(A) $x^2 + 4x + 4$

(B) $x^2 - 4x + 4$

(C) $x^2 - 4$

(D) $(x-4)^2$

(E) $(x+4)^2$

Multiple Choice

14. Given $f(x) = x^2 + a^2$ and $g(x) = x^2 - a^2$ where a is a constant integer. The function $r(x) = \frac{f(x)}{g(x)}$. What is the domain of $r(x)$?

(A) $(-\infty, -a) \cup (a, \infty)$

(B) $(-a, a)$

(C) $(-\infty, -a) \cup (-a, a) \cup (a, \infty)$

(D) $(-\infty, -a)$

(E) (a, ∞)

15. The function $f(x)$ is defined such that for all $x > 0$, the line $y = 4$ is a horizontal asymptote. Which of the following must be true?

(A) $f(4)$ is undefined

(B) $f(x) \neq 4$ for all $x > 0$

(C) $\lim_{x \to \infty} f(x) = 0$

(D) $\lim_{x \to \infty} f(x) = 4$

(E) $f(4) = 0$

16. Given the line $y = a$ is a horizontal asymptote of $g(x)$ for all $x < 0$, which of the following must be true?

(A) $g(a)$ is undefined

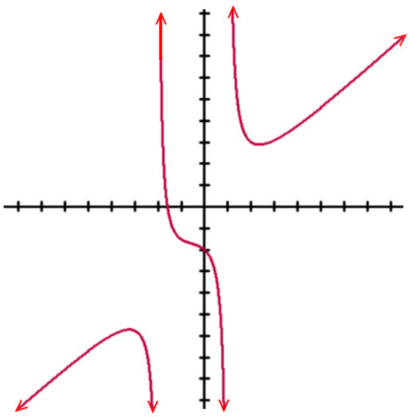
(B) $g(x) \neq a$

(C) $\lim_{x \to -\infty} g(x) = a$

(D) $\lim_{x \to -\infty} g(x) = -a$

(E) None of the above

17. Given the graph of f. Which describes the end behavior of f?



(A) $\lim_{x \to -\infty} f(x) = -\infty$ and $\lim_{x \to \infty} f(x) = -\infty$

(B) $\lim_{x \to -\infty} f(x) = \infty$ and $\lim_{x \to \infty} f(x) = \infty$

(C) $\lim_{x \to -\infty} f(x) = -\infty$ and $\lim_{x \to \infty} f(x) = \infty$

(D) $\lim_{x \to -\infty} f(x) = \infty$ and $\lim_{x \to \infty} f(x) = -\infty$

(E) $\lim_{x \to \infty} f(x) = f(0)$

1.7A Rational Functions and End Behavior

AP Precalculus Name: _____ **CA #1**

State the domain of the following rational functions. Use interval notation.

1. $f(x) = \frac{x^3+1}{x^2-16}$

Domain:

2. $h(x) = \frac{x+3}{2x^2+10x-28}$

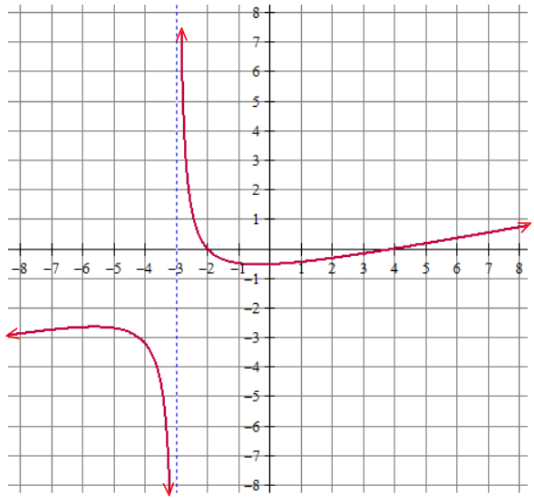
Domain:

3. $d(t) = \frac{t^2-3t+1}{4-3t}$

Domain:

Use the graph of the rational function f to find the following.

4.

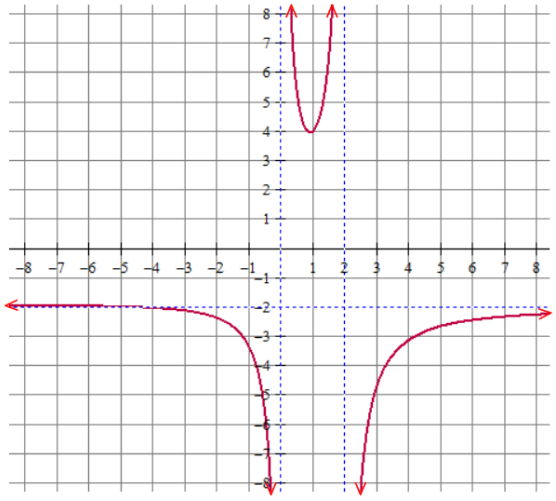


Domain:

End Behavior:

Is there horizontal aysmptote?
If so, write the equation of the horizontal aysmptote.

5.



Domain:

End Behavior:

Is there horizontal aysmptote?
If so, write the equation of the horizontal aysmptote.

Continued on the back.

CALCULATOR ACTIVE Complete the table to answer the following.

6. $f(x) = \frac{x^2}{x^2-4x}$

x	$-10{,}000$	$-1{,}000$	-100	100	$1{,}000$	$10{,}000$
$f(x)$						

End Behavior:

Is there horizontal aysmptote?
If so, write the equation of the horizontal aysmptote.

---

7. $d(t) = \frac{t^3-3t}{2t+1}$

t	$-5{,}000$	-500	-50	50	500	$5{,}000$
$d(t)$						

End Behavior:

Is there horizontal aysmptote?
If so, write the equation of the horizontal aysmptote.

Answers to 1.7A CA #1

1. $(-\infty,-4) \cup (-4,4) \cup (4,\infty)$
2. $(-\infty,-7) \cup (-7,2) \cup (2,\infty)$
3. $\left(-\infty,\frac{4}{3}\right) \cup \left(\frac{4}{3},\infty\right)$
4. Domain: $(-\infty,-3) \cup (-3,\infty)$
   End Behavior:
   $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = \infty$
   Horizontal Asymptote: none
5. Domain: $(-\infty,0) \cup (0,2) \cup (2,\infty)$
   End Behavior:
   $\lim_{x\to-\infty} f(x) = -2$ and $\lim_{x\to\infty} f(x) = -2$
   Horizontal Asymptote: $y = -2$

6.

x	$f(x)$
-10,000	0.9996
-1,000	0.99602
-100	0.9615
100	1.0417
1,000	1.004
10,000	1.0004

$\lim_{x\to-\infty} f(x) = 1$
$\lim_{x\to\infty} f(x) = 1$
HA: $y = 1$

7.

t	$d(t)$
-5,000	$\approx 1.2 \times 10^7$
-500	125124
-50	1261.1
50	1236.1
500	124874
5,000	$\approx 1.2 \times 10^7$

$\lim_{t\to-\infty} f(x) = \infty$
$\lim_{t\to\infty} f(x) = \infty$
HA: none

AP Precalc

Write your questions and thoughts here!

1.7B Rational Functions & End Behavior

1.7B Notes

End Behavior

Example 1:

$f(x) = \dfrac{2x+3}{x^2-4x-32}$

End Behavior:

$\lim\limits_{x\to-\infty} f(x) =$ $\lim\limits_{x\to\infty} f(x) =$

For inputs of large magnitude,
the polynomial in the denominator _____ the polynomial in the numerator.

Example 2:

$g(x) = \dfrac{x^3-3x+5}{5x^2+x+12}$

End Behavior:

Example 3:

$h(t) = \dfrac{2t+3}{t}$

End Behavior:

Finding Horizontal Asymptotes!

Horizontal asymptotes =

- Degree of denominator is bigger:
- Degree of numerator is bigger:
- Degrees are the same:

Degree on bottom is bigger!	Degree on top is bigger!	Degrees are the same!
$f(x) = \dfrac{4x^2+x+8}{x^3+3x^2-1}$	$f(x) = \dfrac{x^2+2x-15}{4x-9}$	$f(x) = \dfrac{x^2+5x-36}{3x^2+21x}$
Horizontal Asymptote:	Horizontal Asymptote:	Horizontal Asymptote:

Write your questions and thoughts here!

Horizontal Asymptote

Let f be the rational function

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_2 x^2 + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + b_{m-2}x^{m-2} + \cdots + b_2 x^2 + b_1 x + b_0}$$

where $N(x)$ and $D(x)$ have no common factors.

a. When $n < m$, the graph of f has the line $y = 0$ as a horizontal asymptote.

b. When $n > m$, the graph of f has no horizontal asymptote.

c. When $n = m$, the graph of f has the line $y = \frac{a_n}{b_n}$ as a horizontal asymptote.

Limits

Evaluate the following limits.

Example 4:

$$\lim_{x\to\infty} \frac{(x+4)(2x-5)}{x^2-9} =$$

Example 5:

$$\lim_{x\to-\infty} \frac{3x^3+2}{x^2-9} =$$

Quick Practice

1. Find the end behavior.

$$d(t) = \frac{t^2 + 2t - 15}{4t - 9}$$

2. Find the horizontal asymptote if one exists.

$$g(x) = \frac{x^4 + 3x^2 - 5}{4x^5 - x^3 + 2x}$$

3. Evaluate the limit.

$$\lim_{n\to-\infty} \frac{3n^5+2}{n^3-9} =$$

1.7B Rational Functions and End Behavior

AP Precalculus

1.7B Practice

Determine the end behavior of the following. Determine the horizontal asymptote if one exists.

1. $f(x) = \frac{x^2-3x+1}{2x+3}$

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

2. $A(r) = \frac{3r^2-4}{5r^3+3r+2}$

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

3. $h(x) = \frac{x^2-9}{4x^2+2x-15}$

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

4. $g(n) = \frac{3n^2-4}{(n+3)(n-7)}$

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

5. $f(x) = \frac{-2x^3+5}{4x^5-8x^3+2x}$

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

6. $g(x) = \frac{-3x^4+x^2+2}{2x^4+7x^2-5}$

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

Find the horizontal asymptote of the following rational function if one exists.

7. $f(x) = \frac{x-1}{x}$

8. $d(t) = \frac{4-t^2}{2t^2+t}$

9. $h(x) = \frac{x^2-1}{4x(x^2+1)}$

Evaluate the following limits.

10. $\lim_{x\to\infty} \frac{3x^2+2}{x^2-9} =$

11. $\lim_{x\to-\infty} \frac{1}{x-4} =$

12. $\lim_{n\to\infty} \frac{2n^5-3n^3+2n}{5n^5+3n-2} =$

13. $\lim_{x\to\infty} \frac{(x+3)(2x-1)}{4x^2-9} =$

14. $\lim_{t\to-\infty} \frac{2t^3-3t}{5t^2-4t} =$

15. $\lim_{n\to-\infty} \frac{n^3}{2n^3-2n} =$

Use the rational function to answer the following.

16. $f(x) = \frac{x^2-2x-24}{2x^3+10x^2}$

a. Domain:

b. y-intercept:

c. $\lim_{x\to-\infty} f(x) =$

d. $\lim_{x\to\infty} f(x) =$

e. As x increases without bound the $f(x)$…

f. As x decreases without bound the $f(x)$…

g. **Multiple Choice** Which of the following is true for input values of large magnitude?

(A) The polynomial in the numerator dominates the polynomial in the denominator indicating no horizontal asymptote.

(B) The polynomial in the numerator dominates the polynomial in the denominator indicating a horizontal asymptote of $y = 0$.

(C) The polynomial of the denominator dominates the polynomial in the numerator indicating no horizontal asymptote.

(D) The polynomial of the denominator dominates the polynomial in the numerator indicating a horizontal asymptote of $y = 0$.

(E) Neither polynomial of the rational function dominates the other indicating a horizontal asymptote of $y = \frac{1}{2}$

17. $g(x) = \frac{4(x+5)(x-2)}{x^2-9}$

a. Domain:

b. y-intercept:

c. $\lim_{x\to-\infty} g(x) =$

d. $\lim_{x\to\infty} g(x) =$

e. As x increases without bound the $g(x)$…

f. As x decreases without bound the $g(x)$…

g. **Multiple Choice** Which of the following is true for input values of large magnitude?

(A) The polynomial in the numerator dominates the polynomial in the denominator indicating no horizontal asymptote.

(B) The polynomial in the numerator dominates the polynomial in the denominator indicating a horizontal asymptote of $y = 0$.

(C) The polynomial of the denominator dominates the polynomial in the numerator indicating no horizontal asymptote.

(D) The polynomial of the denominator dominates the polynomial in the numerator indicating a horizontal asymptote of $y = 0$.

(E) Neither polynomial of the rational function dominates the other indicating a horizontal asymptote of $y = 4$.

Multiple Choice

18. The function f is given by $f(x) = \frac{(x-8)(x+3)}{x^2+5x-24}$. Which of the following describes the function f ?

(A) Domain is $(-\infty,-3) \cup (-3,8) \cup (8,\infty)$ with a horizontal asymptote of $y = 1$.

(B) Domain is $(-\infty,-8) \cup (-8,3) \cup (3,\infty)$ with a horizontal asymptote of $y = 1$.

(C) Domain is $(-\infty,-3) \cup (-3,8) \cup (8,\infty)$ with a horizontal asymptote of $y = 0$.

(D) Domain is $(-\infty,-8) \cup (-8,3) \cup (3,\infty)$with a horizontal asymptote of $y = 0$.

(E) The function does not have a horizontal asymptote.

19. The function f is given by $f(x) = \frac{ax^3-2x^2+5}{2x^3-8}$ and has line $y = 3$ as a horizontal asymptote. Which of the following must be true?

(A) $f(a) = 6$

(B) $a = 6$

(C) $\lim_{x\to\infty} f(x) = a$

(D) $\lim_{x\to\infty} f(x) = 6$

(E) None of the above are true.

20. Which of the following is equivalent to $\lim_{x\to-\infty} \frac{3x^2+2x+6}{5x^4-9x^2+2}$?

(A) $\frac{3}{5}$

(B) $-\frac{3}{5}$

(C) 3

(D) 0

(E) Does not exist.

21. The function f is given by $f(x) = \frac{-x^3+3x^2+x-5}{5x^2+7x-4}$. Which describes the end behavior of f?

(A) $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = -\infty$

(B) $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = \infty$

(C) $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = \infty$

(D) $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = -\infty$

(E) $\lim_{x\to-\infty} f(x) = 0$ and $\lim_{x\to\infty} f(x) = 0$

1.7B Rational Functions and End Behavior

AP Precalculus Name: _____ **CA #1**

Find the end behavior of the following rational functions.		
1. $f(x) = \frac{x+3}{x^2-16}$	2. $g(x) = \frac{5x^2+3x}{2x^2+10x-28}$	3. $h(t) = \frac{t^2-3t+1}{3t+2}$
Find the horizontal asymptote of the following rational functions if one exists.		
4. $f(x) = \frac{(x+2)(x+6)}{2x^2\mp3x-28}$	5. $r(x) = \frac{x+3}{3x^3+5x-2}$	6. $g(n) = \frac{n^2-1}{7+2n}$
Evaluate the following limits.		
7. $\lim_{x\to\infty} \frac{4x^2+2}{x^2-25} =$	8. $\lim_{x\to-\infty} \frac{x^2-4}{x} =$	9. $\lim_{n\to\infty} \frac{n^4-3n^3}{3n^5+n^2-2} =$

Answers to 1.7B CA #1

1. $\lim_{x\to-\infty} f(x) = 0$ $\lim_{x\to\infty} f(x) = 0$	2. $\lim_{x\to-\infty} g(x) = \frac{5}{2}$ $\lim_{x\to\infty} g(x) = \frac{5}{2}$	3. $\lim_{t\to-\infty} h(t) = -\infty$ $\lim_{t\to\infty} h(t) = \infty$
4. $y = \frac{1}{2}$	5. $y = 0$	6. No horizontal asymptote
7. 4	8. $-\infty$ (does not exist)	9. 0

AP Precalc

Write your questions and thoughts here!

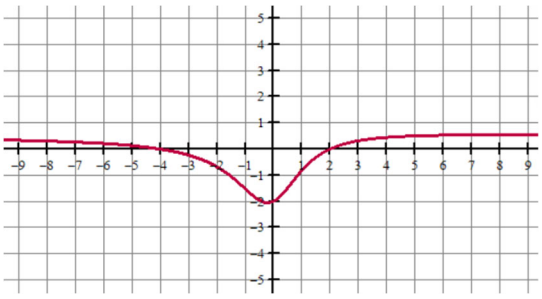
1.8 Rational Functions & Zeros

1.8 Notes

Zeros

Example 1:

$$f(x) = \frac{x^2 + 2x - 8}{2x^2 + 4}$$



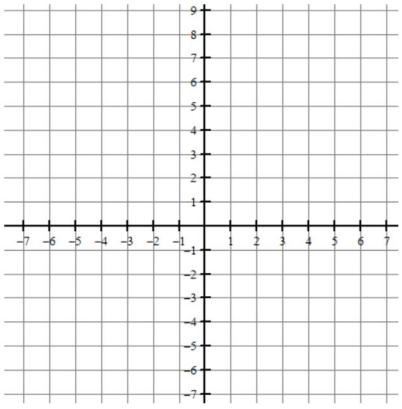
Domain:

Zero(s):

Holes

Example 2:

$$g(x) = \frac{x^2 - x - 12}{x - 4}$$



Domain:

Hole(s):

Zero(s):

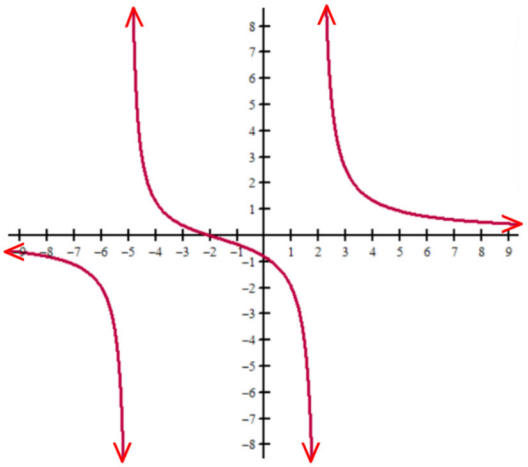
Zeros

Let f be the rational function $f(x) = \frac{N(x)}{D(x)}$ where N and D have no common factors. The zeros of the rational function occur when $N(x) = 0$ for all x in the domain of f.

Vertical Asymptotes

Example 3:

$$f(x) = \frac{4x + 8}{x^2 + 3x - 10}$$



Domain:

Hole(s):

Zero(s):

Vertical Asymptote(s):

Write your questions and thoughts here!

Rational Functions

Example 4:

$$h(x) = \frac{x^2 - 9}{x^2 - 2x - 3}$$

Hole(s):

Zero(s):

Domain:

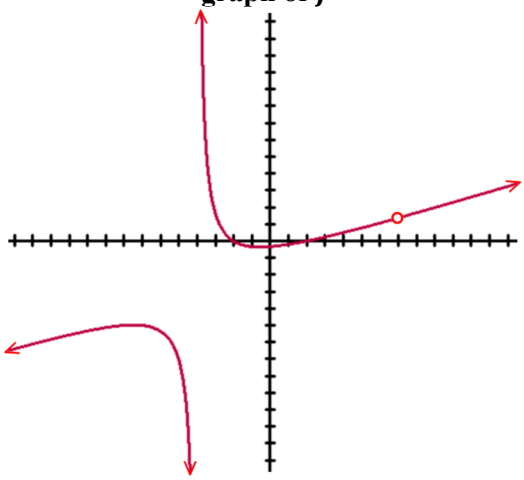
Vertical Asymptote(s):

Horizontal Asymptote:

y-intercept:

Example 5:

graph of *f*



Hole(s):

Zero(s):

Vertical Asymptote(s):

Horizontal Asymptote:

y-intercept (estimate):

Domain:

Sign Table for Example #5

x	$-\infty < x < -4$	-4	$-4 < x < -2$	-2	$-2 < x < 2$	2	$2 < x < 7$	7	$7 < x < \infty$
$f(x)$									

1.8 Rational Functions and Zeros

AP Precalculus

1.8 Practice

Find the zeros of the following rational function if one exists.

1. $f(x) = \frac{x-1}{x^2-9}$

2. $d(t) = \frac{(t+3)(t-1)}{4t+12}$

3. $h(x) = \frac{x^2-3x-10}{x^2+6x}$

4. $r(x) = \frac{x-1}{x}$

5.

x	$-\infty < x < -3$	-3	$-3 < x < 5$	5	$5 < x < \infty$
$f(x)$	Positive	DNE	Negative	0	Positive

6. $c(n) = \frac{n^2+5n}{n^2-25}$

7.

x	$-\infty < x < 1$	1	$1 < x < 6$	6	$6 < x < \infty$
$g(x)$	Negative	0	Negative	0	Positive

Use the rational function to answer the following.

8.

$$f(x) = \frac{x^2 - 2x - 24}{x - 6}$$

a. Domain:

b. Hole(s):

c. Zero(s):

d. Vertical Asymptote(s):

e. Horizontal Asymptote:

f. *y*-intercept:

9.

$$g(x) = \frac{4(x + 5)(x - 2)}{x^2 - 4}$$

a. Domain:

b. Hole(s):

c. Zero(s):

d. Vertical Asymptote(s):

e. Horizontal Asymptote:

f. *y*-intercept:

Use the rational function to answer the following.

10.

$$f(x) = \frac{x+2}{3x^2+6x}$$

a. Domain:

b. Hole(s):

c. x-intercept(s):

d. Vertical Asymptote(s):

e. Horizontal Asymptote:

f. y-intercept:

11.

$$h(t) = \frac{t^3-2t^2}{t^2+3t-18}$$

a. Domain:

b. Hole(s):

c. Root(s):

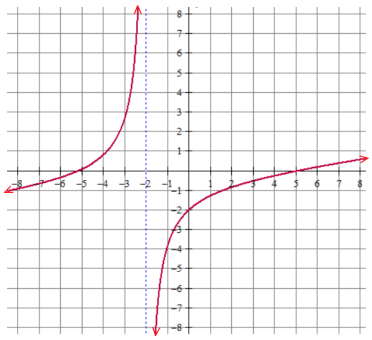
d. Vertical Asymptote(s):

e. Horizontal Asymptote:

f. y-intercept:

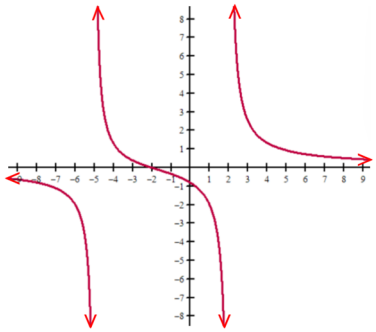
Use the graph to create a sign table.

12.



x	$-\infty < x < -5$	-5	$-5 < x < -2$	-2	$-2 < x < 5$	5	$5 < x < \infty$
$f(x)$							

13.



x	$-\infty < x < -5$	-5	$-5 < x < -2$	-2	$-2 < x < 2$	2	$2 < x < \infty$
$h(x)$							

Multiple Choice

14. The function f is given by $f(x) = \frac{x^2+2x-24}{4-x}$. Which of the following describes the function f ?

(A) The graph of f has an x-intercept at $x = -6$ and a vertical asymptote of $x = 4$.

(B) The graph of f has an x-intercept at $x = -6$ and a hole at $x = 4$.

(C) The graph of f has an x-intercept at $x = -6$ and a vertical asymptote of $x = -4$.

(D) The graph of f has an x-intercept at $x = -6$ and a hole at $x = -4$.

(E) The graph of f has x-intercepts at $x = -6$ and $x = 4$.

For questions 15 and 16 use the following table.

x	$-\infty < x < -3$	-3	$-3 < x < 0$	0	$0 < x < 2$	2	$2 < x < \infty$
$f(x)$	positive	0	negative	undefined	negative	0	positive

15. Which of the following must be true for the function f ?

(A) The graph of f has a maximum at $x = -3$ and a minimum at $x = 2$.

(B) The graph of f has a minimum at $x = -3$ and a maximum at $x = 2$.

(C) f has exactly two distinct real zeros.

(D) f has exactly three distinct real zeros.

(E) The graph of f has a vertical asymptote at $x = 0$.

16. Which of the following could be an expression for $f(x)$?

(A) $\frac{x(x+3)(x-2)}{x}$

(B) $\frac{x(x-3)(x+2)}{x}$

(C) $\frac{x}{x(x+3)(x-2)}$

(D) $\frac{x}{x(x-3)(x+2)}$

(E) None of the above

1.8 Rational Functions and Zeros

AP Precalculus Name: _____ **CA #1**

Find the zeros of the following rational functions.

1. $f(x) = \dfrac{x+2}{x^2-9}$

2. $g(x) = \dfrac{x^2+3x}{x^2-4x-21}$

3. $h(t) = \dfrac{t^2-3t+2}{t-2}$

Use the rational functions to answer the following

4. $f(x) = \dfrac{(x+3)(x+5)}{x^2+3x}$

a. Domain:

b. Hole(s):

c. Zero(s):

d. Vertical Asymptote(s):

5. $d(t) = \dfrac{t^2-1}{t+3}$

a. Domain:

b. Hole(s):

c. Zero(s):

d. Vertical Asymptote(s):

6. $r(x) = \dfrac{x^2-3x-18}{2x-12}$

a. Domain:

b. Hole(s):

c. Zero(s):

d. Vertical Asymptote(s):

Answers to 1.8 CA #1

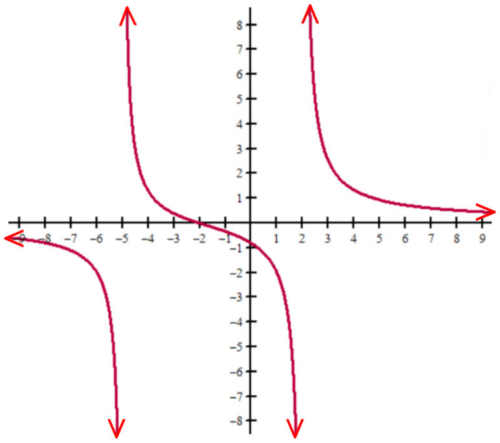
1. $x = -2$	2. $x = 0$	3. $t = 1$
4. a. $(-\infty,-3) \cup (-3,0) \cup (0,\infty)$ b. $x = -3$ c. $x = -5$ d. $x = 0$	5. a. $(-\infty,-3) \cup (-3,\infty)$ b. No holes c. $t = -1$ and 1 d. $t = -3$	6. a. $(-\infty,6) \cup (6,\infty)$ b. $x = 6$ c. $x = -3$ d. No Vertical Asymptotes

1.9 Rational Functions & Vertical Asymptotes

1.9 Notes

Vertical Asymptotes

Example 1:



Vertical Asymptote(s):

Limit Notation:

$\lim_{x \to -5^-} f(x) =$

As x approaches -5 from the left the $f(x)$…

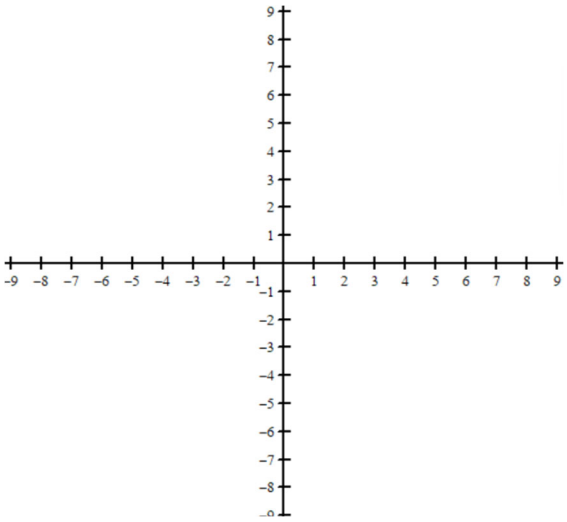
$\lim_{x \to -5^+} f(x) =$

As x approaches -5 from the right the $f(x)$…

Example 2:

$$g(x) = \frac{x^2 - 9}{x^2 - 4}$$

Limit Notation of Vertical Asymptote(s):



Vertical Asymptotes

Let f be the rational function

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_2 x^2 + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + b_{m-2}x^{m-2} + \cdots + b_2 x^2 + b_1 x + b_0}$$

where $N(x)$ and $D(x)$ have no common factors.

The graph of f has vertical asymptotes at the zeros of $D(x)$.

Write your questions and thoughts here!

Numerically

Example 3:

$$g(x) = \frac{x^2 + 3x}{x - 2}$$

As x approaches 2 from the left the $g(x)$…

x	$g(x)$
1.9	
1.99	
1.999	
1.9999	

As x approaches 2 from the right the $g(x)$…

x	$g(x)$
2.1	
2.01	
2.001	
2.0001	

Multiplicity

Example 4:

$$h(x) = \frac{x^2 + 4x - 12}{(x - 2)^2}$$

Domain:

Vertical Asymptote(s):

Limit Notation of Vertical Asymptote(s):

Hole(s):

Zero(s):

Horizontal Asymptote:

End Behavior:

1.9 Rational Functions and Vertical Asymptotes

AP Precalculus

1.9 Practice

Find the domain and vertical asymptote(s) of the following rational function if one exists.

1. $f(x) = \frac{x(x+2)}{x^2-4}$

Domain:

Vertical Asymptote(s):

2. $d(t) = \frac{t^2+4t-12}{(t-2)^2}$

Domain:

Vertical Asymptote(s):

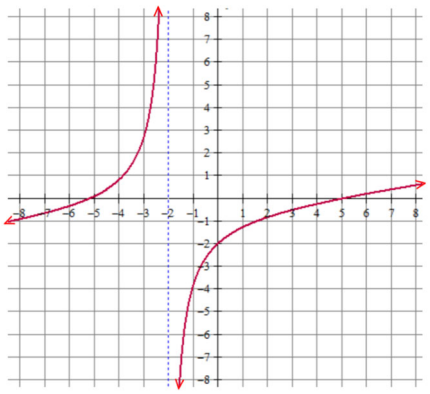
3. $h(x) = \frac{x^3-3x^2}{x^2+8x+15}$

Domain:

Vertical Asymptote(s):

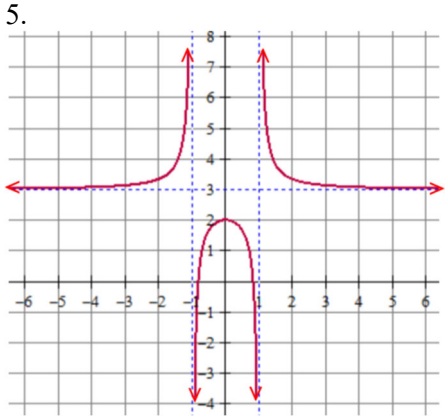
State the vertical asymptote(s) of the following rational functions. Use limit notation.

4.



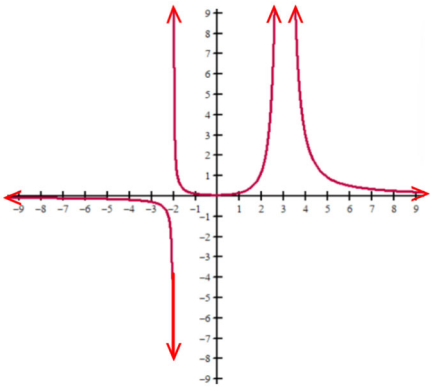
Limit Notation Vertical Asymptote(s):

5.



Limit Notation Vertical Asymptote(s):

6.



Limit Notation Vertical Asymptote(s):

CALCULATOR ACTIVE Complete the table to answer the following.

7. $f(x) = \frac{x^2-1}{x-4}$

x	3.9	3.99	3.999	4	4.001	4.01	4.1
$f(x)$							

Vertical Asymptote:

Limit Notation of Vertical Asymptote:

CALCULATOR ACTIVE Complete the table to answer the following.

8. $f(x) = \frac{x^2-2x}{x+2}$

x	-2.1	-2.01	-2.001	-2	-1.999	-1.99	-1.9
$f(x)$							

Vertical Asymptote: Limit Notation of Vertical Asymptote:

Use the table of the rational function *h* to find the following.

9.

t	$d(t)$
-0.1	5,589
-0.01	37,231
-0.001	96,543
-0.0001	148,234
0	undefined
0.0001	128,341
0.001	89,437
0.01	18,235
0.1	1,455

a. Find $d(0) =$

b. Find the *y*-intercept.

c. Find $\lim_{t\to 0^-} d(t) =$

d. Find $\lim_{t\to 0^+} d(t) =$

e. As *t* approaches zero from the left the $d(t)$…

f. As *t* approaches zero from the right the $d(t)$…

Make a sketch of the rational function with the following characteristics.

10. The graph of f has…

a. $f(-4) = 0$

b. $f(6) = 0$

c. $\lim_{x\to -3^-} f(x) = -\infty$

d. $\lim_{x\to -3^+} f(x) = \infty$

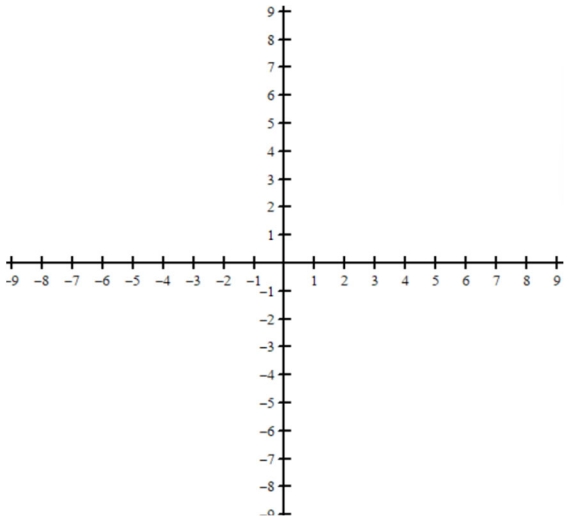
e. $\lim_{x\to 4^-} f(x) = \infty$

f. $\lim_{x\to 4^+} f(x) = -\infty$

g. $\lim_{x\to -\infty} f(x) = 2$

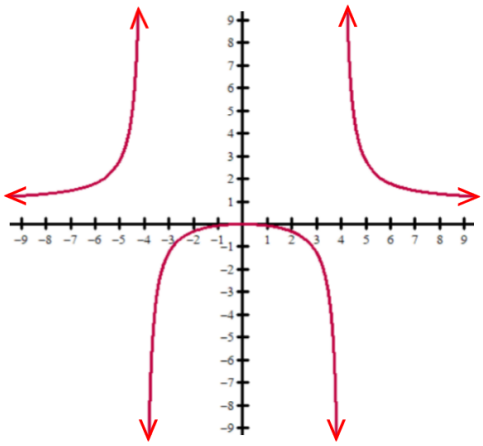
h. $\lim_{x\to \infty} f(x) = 2$

i. $f(0) = 5$



Multiple Choice

11. Given the graph of f. Which of the following describes the function f?



(A) $\lim_{x\to -4^-} f(x) = -\infty$ and $\lim_{x\to -4^+} f(x) = -\infty$

(B) $\lim_{x\to -4^-} f(x) = \infty$ and $\lim_{x\to -4^+} f(x) = -\infty$

(C) $\lim_{x\to -4^-} f(x) = -\infty$ and $\lim_{x\to -4^+} f(x) = \infty$

(D) $\lim_{x\to -4^-} f(x) = \infty$ and $\lim_{x\to -4^+} f(x) = \infty$

(E) $\lim_{x\to -4} f(x) = f(0)$

Free Response

12. The function f is a rational function graphed in the *xy*-plane. The polynomial in the numerator of f has exactly one real zero at $x = 3$. The polynomial of the denominator of f has exactly two real zeros at both $x = 3$ and $x = 6$. The multiplicities of the zeros at $x = 3$ in the numerator and in the denominator are equal.

    a. Find the domain for the graph of f.

    b. Describe any holes and/or vertical asymptotes for the graph of f.

    c. Explain how your answer from part b would change if the multiplicities of the zeros at $x = 3$ in the numerator and denominator were not equal?

1.9 Rational Functions & Vertical Asymptotes

AP Precalculus Name: _____ **CA #1**

Find the vertical asymptote(s) of the following rational functions, if one exists.

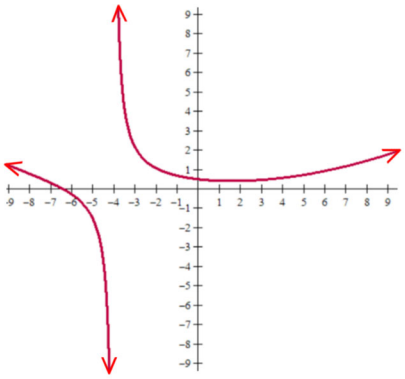
1. $f(x) = \dfrac{x^2 - 4}{x^2 - 8x - 20}$

2. $g(x) = \dfrac{x^2 + 3x}{x^2 - 9x + 18}$

3. $h(t) = \dfrac{t^2 - 2t - 48}{t^2 + 12t + 36}$

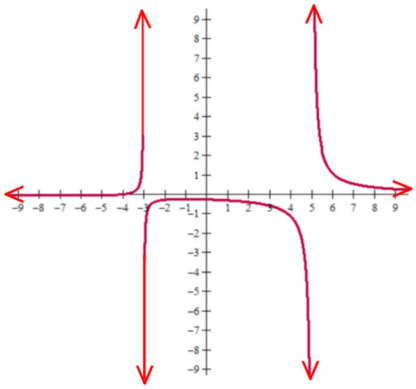
Use the graph of f to find the vertical asymptote(s). Use limit notation.

4.



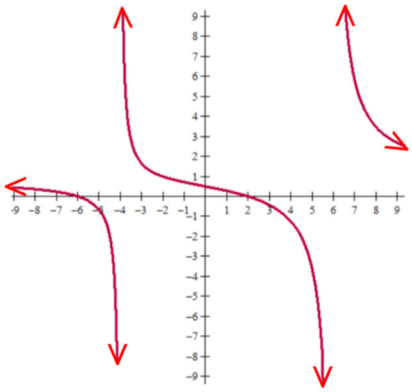
Limit Notation Vertical Asymptote(s):

5.



Limit Notation Vertical Asymptote(s):

6.



Limit Notation Vertical Asymptote(s):

CALCULATOR ACTIVE Complete the table to answer the following.

7. $f(x) = \frac{x+5}{x-2}$

x	1.9	1.99	1.999	2	2.001	2.01	2.1
$f(x)$							

Vertical Asymptote: Limit Notation of Vertical Asymptote:

CALCULATOR ACTIVE Complete the table to answer the following.

8. $f(x) = \frac{x^3+8}{x^2+3x}$

x	-3.1	-3.01	-3.001	-3	-2.999	-2.99	-2.9
$f(x)$							

Vertical Asymptote: Limit Notation of Vertical Asymptote:

Answers to 1.9 CA #1

1. $x = 10$

2. $x = 3$ and 6

3. $t = -6$

4. $\lim_{x\to -4^-} f(x) = -\infty$ and $\lim_{x\to -4^+} f(x) = \infty$

5. $\lim_{x\to -3^-} f(x) = \infty$ and $\lim_{x\to -3^+} f(x) = -\infty$
$\lim_{x\to 5^-} f(x) = -\infty$ and $\lim_{x\to 5^+} f(x) = \infty$

6. $\lim_{x\to -4^-} f(x) = -\infty$ and $\lim_{x\to -4^+} f(x) = \infty$
$\lim_{x\to 6^-} f(x) = -\infty$ and $\lim_{x\to 6^+} f(x) = \infty$

7.

x	$f(x)$
1.9	-69
1.99	-699
1.999	-6999
2	undefined
2.001	7001
2.01	701
2.1	71

vertical asymptote: $x = 2$

$\lim_{x\to 2^-} f(x) = -\infty$ $\lim_{x\to 2^+} f(x) = \infty$

8.

x	$f(x)$
-3.1	-70.29
-3.01	-640.2
-3.001	-6340
-3	undefined
-2.999	6326.4
-2.99	626.45
-2.9	56.51

vertical asymptote: $x = -3$

$\lim_{x\to -3^-} f(x) = -\infty$ $\lim_{x\to -3^+} f(x) = \infty$

AP Precalc

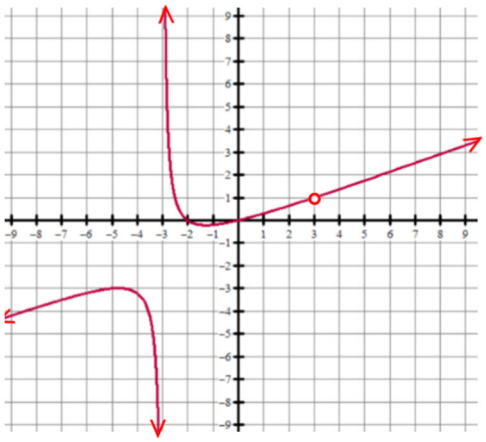
1.10 Rational Functions and Holes

1.10 Notes

Write your questions and thoughts here!

Holes

Example 1:



Hole(s):

Limit Notation:

$\lim_{x \to 3^-} f(x) =$

As x approaches 3 from the left the $f(x)$…

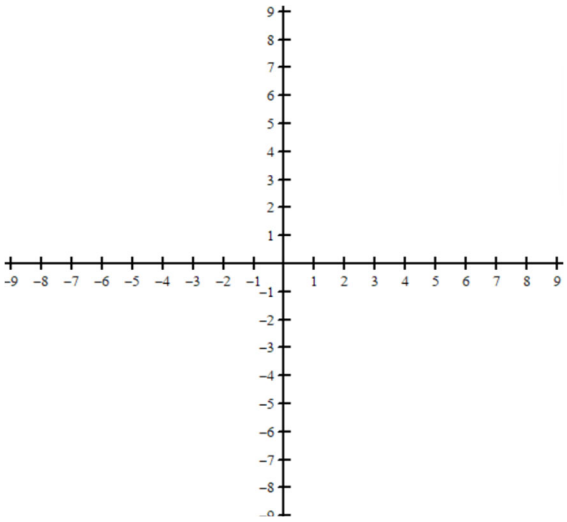
$\lim_{x \to 3^+} f(x) =$

As x approaches 3 from the right the $f(x)$…

Example 2:

$$g(x) = \frac{10x + 30}{x^2 + x - 6}$$

Limit Notation of Hole(s):



Numerically

As x approaches -3 from the left the $g(x)$…

x	$g(x)$
-3.1	
-3.01	
-3.001	
-3.0001	

As x approaches -3 from the right the $g(x)$…

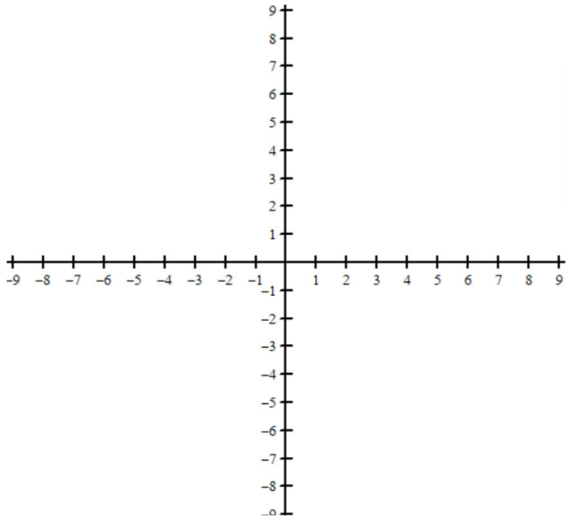
x	$g(x)$
-2.9	
-2.99	
-2.999	
-2.9999	

Write your questions and thoughts here!

Example 3:

$$g(x) = \frac{(-x^2 + 2x + 4)(x - 1)}{x - 1}$$

Can look like polynomials!



Holes

Let f be the rational function $f(x) = \frac{N(x)}{D(x)}$. The holes of the rational function occur at all common factors whose multiplicity in the numerator is greater than or equal to the the multiplicity in the denominator.

Multiplicity

Example 4:

$$h(x) = \frac{(x - 2)^2}{x^2 + 4x - 12}$$

Domain:

Vertical Asymptote(s):

Hole(s):

Horizontal Asymptote:

Zero(s):

End Behavior:

1.10 Rational Functions and Holes

AP Precalculus

1.10 Practice

Find the hole(s) of the following rational function if one exists.

1. $f(x) = \frac{x^2+3x}{x^2-9}$

Hole(s):

2. $d(t) = \frac{t^2+t-20}{t+5}$

Hole(s):

3. $h(x) = \frac{(x+5)^2}{x^2+8x+15}$

Hole(s):

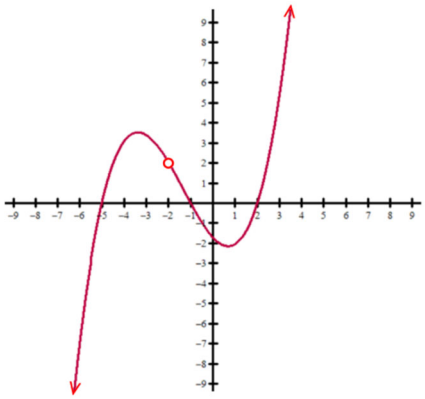
State the hole(s) of the following rational functions. Use limit notation.

4.



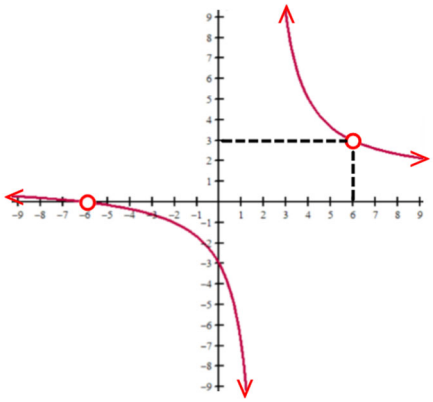
Limit Notation Hole(s):

5.



Limit Notation Hole(s):

6.



Limit Notation Hole(s):

CALCULATOR ACTIVE Complete the table to answer the following.

7. $f(x) = \frac{x^2-16}{x-4}$

x	3.9	3.99	3.999	4	4.001	4.01	4.1
$f(x)$							

Hole: Limit Notation of Hole:

CALCULATOR ACTIVE Complete the table to answer the following.

8. $f(x) = \frac{x^2-2x-8}{4x+8}$

x	-2.1	-2.01	-2.001	-2	-1.999	-1.99	-1.9
$f(x)$							

Hole: Limit Notation of Hole:

Use the table of the rational function *d* to find the following.

9.

t	$d(t)$
-3.1	4.134
-3.01	4.15
-3.001	4.1893
-3.0001	4.1998
-3	undefined
-2.9999	4.2014
-2.999	4.231
-2.99	4.305
-2.9	4.37

a. Find $\lim_{t\to-3^-} d(t) =$

b. Find $\lim_{t\to-3^+} d(t) =$

c. As *t* approaches negative three from the left the $d(t)$…

d. As *t* approaches negative three from the right the $d(t)$…

Use the graph of the rational function *f* to find the following.

10.

a. $f(6) =$

b. $f(2) =$

c. $\lim_{x\to-5^-} f(x) =$

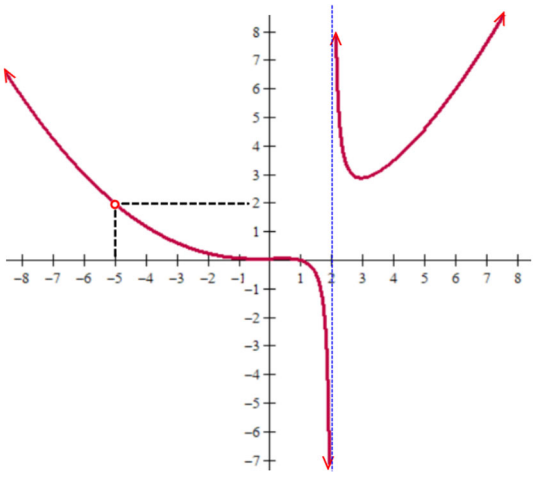
d. $\lim_{x\to-5^+} f(x) =$

e. $\lim_{x\to2^-} f(x) =$

f. $\lim_{x\to2^+} f(x) =$

g. $\lim_{x\to-\infty} f(x) =$

h. $\lim_{x\to\infty} f(x) =$



i. Domain =

Multiple Choice

11. The function f is given by $f(x) = \frac{x^2-9}{x^2+8x+15}$. Which of the following describes the function f ?

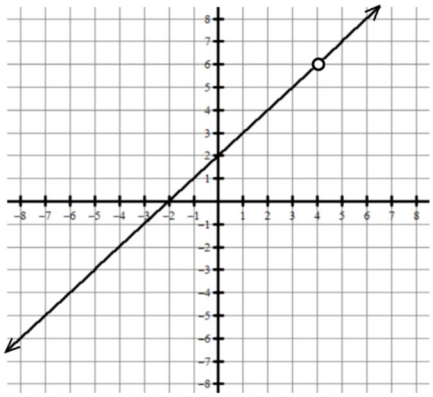
(A) There is a hole at $x = 5$.

(B) There is a hole at $x = -5$.

(C) There is a hole at $x = 3$.

(D) There is a hole at $x = -3$.

For questions 12-13 use the graph of f.



graph of f

12. The figure shows the graph of a function f. Which of the following could be an expression for the $f(x)$?

(A) $\frac{(x+2)(x-4)}{(x+2)}$

(B) $\frac{(x-2)(x+4)}{(x-2)}$

(C) $\frac{(x+2)(x-4)}{(x-4)}$

(D) $\frac{(x-2)(x+4)}{(x+4)}$

13. The figure shows the graph of a function f. Which of the following must be true?

(A) $f(2) = 0$

(B) $\lim_{x\to 4^+} f(x) = \infty$

(C) $\lim_{x\to 4^+} f(x) = 3f(0)$

(D) $f(-2) + f(0) = 0$

1.10 Rational Functions & Holes

AP Precalculus

Name: _____

CA #1

Find the hole(s) of the following rational functions, if one exists.

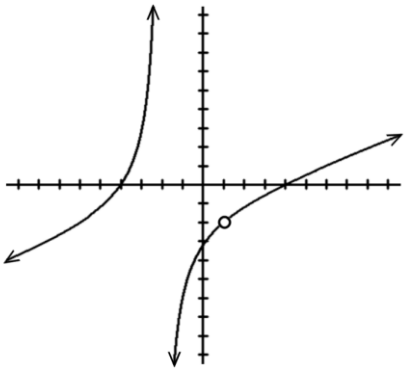
1. $$f(x) = \frac{x^2 - 36}{x^2 - 8x + 12}$$

2. $$g(x) = \frac{5x^2 + 20x}{x^2 + 9x + 20}$$

3. $$h(t) = \frac{t^2 + 12t + 36}{t^2 + t - 30}$$

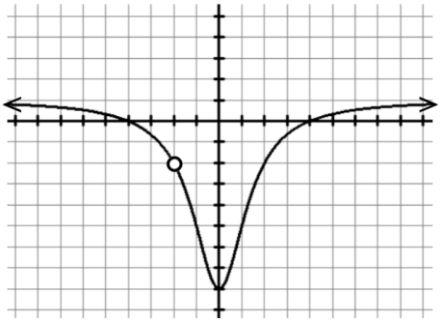
Use the graph of f to find the hole(s). Use limit notation.

4.



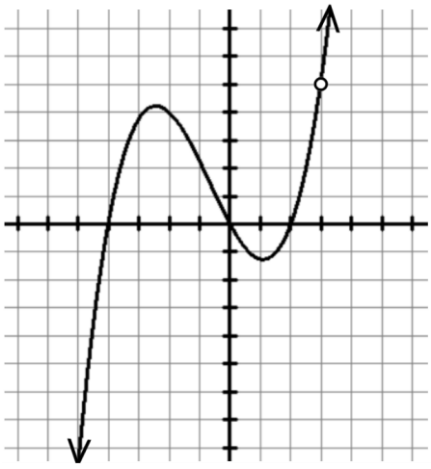
Limit Notation Hole(s):

5.



Limit Notation Hole(s):

6.



Limit Notation Hole(s):

CALCULATOR ACTIVE Complete the table to answer the following.

7. $f(x) = \frac{x^2+3x-10}{x^2-2x}$

x	1.9	1.99	1.999	2	2.001	2.01	2.1
$f(x)$							

Hole: Limit Notation of Hole:

CALCULATOR ACTIVE Complete the table to answer the following.

8. $f(x) = \frac{x^3+8}{x+2}$

x	-2.1	-2.01	-2.001	-2	-1.999	-1.99	-1.9
$f(x)$							

Hole: Limit Notation of Hole:

Answers to 1.10 CA #1

1. $x = 6$

2. $x = -4$

3. $t = -6$

4. $\lim_{x\to 1^-} f(x) = -2$ and $\lim_{x\to 1^+} f(x) = -2$

5. $\lim_{x\to -2^-} f(x) = -2$ and $\lim_{x\to -2^+} f(x) =$

6. $\lim_{x\to 3^-} f(x) = 5$ and $\lim_{x\to 3^+} f(x) = 5$

7.

x	$f(x)$
1.9	3.6316
1.99	3.5126
1.999	3.5013
2	undefined
2.001	3.4999
2.01	3.4876
2.1	3.381

hole: $x = 2$

$\lim_{x\to 2^-} f(x) = 3.5$ $\lim_{x\to 2^+} f(x) = 3.5$

8.

x	$f(x)$
-2.1	12.61
-2.01	12.06
-2.001	12.006
-2	undefined
-1.999	11.994
-1.99	11.94
-1.9	11.41

hole: $x = -2$

$\lim_{x\to -2^-} f(x) = 12$ $\lim_{x\to -2^+} f(x) = 12$

AP Precalc

Write your questions and thoughts here!

1.11A Equivalent Representations & Binomial Theorem

1.11A Notes

Factored Form VS General Form

POLYNOMIAL FUNCTION

Factored Form

$f(x) = (x+5)^3(x-2)^2$

Standard Form

$f(x) = 3x^4 + 2x^3 - 8x + 5$

RATIONAL FUNCTION

Factored Form

$f(x) = \frac{(x+4)(x-5)}{(x+4)(x-4)}$

General Form

$f(x) = \frac{4x-8}{x^2-2x-2}$

Example #1

$f(x) = 2x^3 - 16x^2 + 32x$

Find…

a. Degree:

b. End Behavior:

c. *y*-intercept:

d. Zero(s):

e. Where is $f(x) \geq 0$?

Example #2

$f(x) = \frac{3x(x-5)}{(x+3)(x-1)}$

Find…

a. Domain:

b. Zero(s):

c. Hole(s):

d. Vertical Asymptote(s):

e. Horizontal Asymptote:

f. *y*-intercept:

Write your questions and thoughts here!

Binomial Theorem

$f(x) = (x+5)^2$

$f(x) = (x+5)^3$

$f(x) = (x+5)^4$

Pascal's Triangle

Example 3:
Expand the following.

$$f(x) = (x-3)^5$$

Example 4:
Find the third term in the expansion.

$$f(x) = (2x+3)^6$$

Binomial Theorem

The expansion of a binomial is…

$$(a+b)^n = C_1 a^n b^0 + C_2 a^{n-1} b^1 + C_3 a^{n-2} b^2 + \cdots + C_{n-1} a^1 b^{n-1} + C_n a^0 b^n$$

Where C is the coefficient given by Pascal's Triangle.

1.11A Equivalent Representations and Binomial Theorem

AP Precalculus

1.11A Practice

The following are in factored form. Convert to standard form and answer the questions.

1. $f(x) = 2(x-4)^2$

   a. Standard Form:

   b. Degree:

   c. End Behavior:

   d. Zero(s):

   e. Where is $f(x) \geq 0$?

   f. *y*-intercept:

2. $h(x) = \frac{2(x+3)(x-3)}{(x+2)(x+3)}$

   a. General Form:

   b. Domain:

   c. Zero(s):

   d. Hole(s):

   e. Vertical Asymptote(s):

   f. Horizontal Asymptote:

   g. *y*-intercept:

The following are in standard form. Convert to factored form and answer the questions.

3. $g(x) = -3x^3 + 27x$

   a. Factored Form:

   b. Degree:

   c. End Behavior:

   d. Zero(s):

   e. Where is $f(x) \geq 0$?

   f. *y*-intercept:

4. $h(t) = \frac{t^3-4t^2}{t^2+2t-15}$

   a. Factored Form:

   b. Domain:

   c. Zero(s):

   d. Hole(s):

   e. Vertical Asymptote(s):

   f. Horizontal Asymptote:

   g. *y*-intercept:

Use the binomial theorem to expand the following.

5. $(x+2)^4$

6. $(x-3)^5$

7. $(2x+1)^3$

8. $(3x-2)^4$

Find the given term in the binomial expansion.

9. $(x+5)^5$

Find the 3rd term.

10. $(x-2)^7$

Find the 4th term.

11. $(3x-1)^8$

Find the 5th term.

12. $(2x+3)^4$

Find the last term.

Multiple Choice

13. What is the leading coefficient of the fourth term when $(x-2)^6$ is expanded?

(A) -160

(B) -120

(C) 120

(D) 160

14. What is the second term when $(2a-b)^4$ is expanded?

(A) $-32a^3b$

(B) $-4a^2b^2$

(C) $-16ab^3$

(D) $-ab^4$

15. In the xy-plane, the graph of a function f has $\lim_{x\to\infty} f(x) = \infty$. Which of the following could be an expression $f(x)$?

(A) $\frac{(x+3)^3}{x(x^3+1)}$

(B) $-2(x-1)^3(x+5)$

(C) $\frac{x(x+3)^3}{x(x^3+1)}$

(D) $x(x+2)^3(x-1)$

1.11A Equivalent Expressions and Binomial Theorem

AP Precalculus Name: _____ **CA #1**

Convert to standard or general form and answer the questions.

1. $f(x) = 2(x+2)^2$

   a. Standard Form:

   b. Degree:

   c. *y*-intercept:

2. $h(t) = \dfrac{(t+1)(t-5)}{3(t^3+1)}$

   a. General Form:

   b. Horizontal Asymptote:

   c. *y*-intercept:

Convert to factored form and answer the questions.

3. $f(x) = 2x^3 - 18x$

   a. Factored Form:

   b. Zero(s):

   c. Where is $f(x) \geq 0$?

4. $g(x) = \dfrac{2x-6}{x^2-9}$

   a. Factored Form:

   b. Zero(s):

   c. Vertical Asymptote(s):

Use the binomial theorem to expand the following.

5. $(x-5)^4$

6. $(2x+1)^5$